

HW 7

Exercise 1 (Data set: `X.csv`).

- a. Fit the model $E(y) = \beta_0 + \beta_1 x$ to the data.
- b. Calculate the residuals for the model.
- c. Plot the residuals versus x . Do you detect any trends? If so, what does the pattern suggest about the model?
- d. Plot the residuals for the first-order model versus $\hat{y}$. Do you detect any trends? If so, what does the pattern suggest about the model?

Exercise 2 (Data set: `R.csv`).

Consider the dataset `R.csv` and build a 1st order MLR model with all variables.

- a. Plot the model residuals against each of the four independent variables using R.
- b. How would you advise the researchers to modify the model? Explain